

Practical 1

Practice Lab Assignment:

CODETANTRA

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1.1.1. Calculate Momentum

05:15

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 mass=float(input())
2 velo=float(input())
3 mom=mass*velo
4 print(f"{mom:.2f}kgm/s")
5
```

Average time
0.015 s
15.25 ms

Maximum time
0.019 s
19.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 19 ms

Expected output
5.0
10.0
50.00kgm/s

Actual output
5.0
10.0
50.00kgm/s

Test case 2 14 ms

Terminal

Test cases

< Prev

Reset

Submit

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1.1.2. Conditional Calculation Based on the Number of Digits

13:55



Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints: $1 \leq n \leq 999$ **Input Format:**

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases



condition...



Submit



```
1 #write your code here...
2 import math
3 n=int(input())
4 if n>=1 and n<=9:
5     print(n*n)
```

Average time

0.006 s

6.00 ms



Maximum time

0.007 s

7.00 ms



4 out of 4 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 6 ms

Debug



Expected output

Actual output

9

9

81

81

Test case 2 5 ms



Test case 3 6 ms



Terminal

Test cases

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1.1.3. Days between Two Dates

16:15



Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format *YYYY – MM – DD*.
- The second line contains the second date in the format *YYYY – MM – DD*.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases



noOfDay...



Submit



```
1 from datetime import date
2 d1=input()
3 d2=input()
4 y1,m1,day1=map(int,d1.split('-'))
5 y2,m2,day2=map(int,d2.split('-'))
```

Average time

0.007 s

7.00 ms



Maximum time

0.009 s

9.00 ms



2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 9 ms

Debug



Expected output

Actual output

2014-07-02

2014-07-02

2014-11-02

2014-11-02

123

123

Test case 2 5 ms



Terminal

Test cases

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Submit

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1.1.4. Reverse a Number

01:31



Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases



reverseN...



Submit

Debugger

```
1 #write your code here...
2 n=int(input())
3 rev=int(str(n)[::-1])
4 print(rev)
```

Average time

0.004 s

4.40 ms



Maximum time

0.006 s

6.00 ms



2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 5 ms

Debug



Expected output

5367

7635

Actual output

5367

7635

Test case 2 4 ms



Terminal

Test cases

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1.1.5. Multiplication Table

02:00



Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases



multiplica...



Submit



```
1 #write your code here...
2 n=int(input())
3 for i in range(1,11,1):
4     print(f"{n} x {i} = {n*i}")
```

Average time

0.007 s



7.00 ms

Maximum time

0.012 s



12.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 12 ms

Debug



Expected output

Actual output

8

8

8 x 1 = 8

8 x 1 = 8

8 x 2 = 16

8 x 2 = 16

8 x 3 = 24

8 x 3 = 24

8 x 4 = 32

8 x 4 = 32

8 x 5 = 40

8 x 5 = 40

Terminal

Test cases

< Prev

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Lab Assignment:

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1.2.1. Pass or Fail

14:12

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

passorFa...

Submit

```
1 n=int(input())
2 marks= list(map(int, input().split()))
3 for m in marks:
4     if m < 40:
5         print("Fail")
```

Average time
0.009 s
8.70 ms

Maximum time
0.013 s
13.00 ms

5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed

Test case 1 13 ms Debug

Expected output	Actual output
4	4
55 65 78 89	55 65 78 89
Aggregate Percentage: 71.75	Aggregate Percentage: 71.75
Grade: First Division	Grade: First Division

Test case 2 10 ms

Terminal

Test cases

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Reset

Submit

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1.2.2. Fibonacci Series

10:18

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

+

fibonacci...

Submit

```
1 def fibonacci(n):  
2  
3     if n==1:  
4         return 0  
5     elif n==2:
```

Average time

0.017 s

17.00 ms

Maximum time

0.042 s

42.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 7 ms

Debug

Expected output

5

0 1 1 2 3

Actual output

5

0 1 1 2 3

Test case 2 10 ms

Terminal

Test cases

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1.2.3. Pattern - 1

02:45

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

+

rightangl...

Submit

```
1 #Type Content here...
2 rows = int(input())
3 for i in range(1, rows + 1):
4     for j in range(i):
5         print("* ", end="")
```

Average time

0.007 s

7.33 ms

Maximum time

0.010 s

10.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 7 ms

Debug

^

Expected output

5

* .

* * .

* * * .

* * * * .

* * * * * .

Actual output

5

* .

* * .

* * * .

* * * * .

* * * * * .

Terminal

Test cases

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1.2.4. Pattern - 2

04:25



Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases



numberP...



Submit

```
1 # Type Content here...
2 rows = int(input())
3 for i in range(1, rows + 1):
4     for j in range(1, i + 1):
5         print(j, end=" ")
```

Average time

0.005 s

5.25 ms

Maximum time

0.008 s

8.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 8 ms

Debug



Expected output

Actual output

5

5

1

1

1 2

1 2

1 2 3

1 2 3

1 2 3 4

1 2 3 4

1 2 3 4 5

1 2 3 4 5

Terminal

Test cases

< Prev

Reset

Submit

Next >